

The cradle of the deltaF508 mutation.

PMID: 19999232 | Indexed in PubMed

Cystic fibrosis (CF) is the most common autosomal recessive disorder caused due to mutation/s in the CFTR gene. The most common mutation in CFTR worldwide is deltaF508 and cystic fibrosis genetic analysis consortium revealed that this mutation is responsible for approximately 66% of all CF chromosomes in the world. Studies looking at the DNA polymorphic haplotypes created by CF linked markers suggest that deltaF508 has a single origin as this mutation has been found associated exclusively with one marker haplotype. Despite a high prevalence of this mutation in CF patients in northern parts of Europe, findings suggest that this mutation was not spread by Europeans but by a group that is speculated to have originated in the Middle East or a more eastern region in Asia (most likely subcontinent). Over here we have given a brief introduction to cystic fibrosis and classification of CFTR mutations and have further elaborated on the crucial issue about the spread of the deltaF508 mutation. We have reviewed findings that give clues about the origin of this mutation from the Baluch ethnicity residing in Pakistan.

Efficacy of lumacaftor-ivacaftor for the treatment of cystic fibrosis patients homozygous for the F508del-CFTR mutation.

PMID: 27482545 | Indexed in PubMed

Cystic fibrosis (CF) results from mutations in the CF transmembrane conductance regulator (CFTR) gene, which codes for the CFTR channel protein. The most common mutation in CF is F508del, which produces a misfolded protein with diminished channel activity. The development of small-molecule CFTR-modulator compounds offers an exciting and novel approach for pharmacological treatment of CF. The corrector lumacaftor helps rescue F508del-CFTR to the cell surface, and potentiator ivacaftor increases F508del-CFTR channel activity. The combination of lumacaftor-ivacaftor (Vertex Pharmaceuticals Incorporated) represents the first FDA-approved therapy for CF patients with two copies of the F508del mutation. Although this combination therapy is the first treatment to directly target the F508del-CFTR mutation, patients taking this drug displayed only modest improvements in lung function. This article summarizes recent data from clinical trials and research discoveries relating to the lumacaftor-ivacaftor treatment, and considers options for identifying future therapies that will be most efficacious for all CF patients.

p.G970D is the most frequent CFTR mutation in Chinese patients with cystic fibrosis.

PMID: 27081564 | Indexed in PubMed

Cystic fibrosis (CF), the most common life-threatening autosomal recessive disorder in Caucasians, is caused by mutations in CF transmembrane conductance regulator gene (CFTR). We and others previously identified CFTR mutations in 20 Chinese patients with CF. In this study, eight Chinese patients with a clinical diagnosis of suspected CF were newly collected and screened for CFTR mutations using a combination of conventional Sanger sequencing and multiplex ligation-dependent probe amplification (MLPA) analysis. The CFTR mutations observed in Chinese CF patients, both reported previously and identified in the present study, were also summarized. In the newly collected patients, we identified 10

different CFTR mutations, including p.F508del, the most common CF-causing mutation in Caucasians, and three novel mutations (p.V1212Afs*15; p.L666* and p.A969A). Most notably, the previously reported p.G970D mutation was found in six patients, making it the most frequent CFTR mutation identified in Chinese CF patients thus far. In conclusion, we detected p.F508del for the first time, identified additional novel CFTR mutations and recorded the most frequent CF-causing mutation in Chinese CF patients.

Cystic fibrosis transmembrane conductance regulator (CFTR) allelic variants relate to shifts in faecal microbiota of cystic fibrosis patients.

PMID: 23613805 | Indexed in PubMed

In this study we investigated the effects of the Cystic Fibrosis Transmembrane conductance Regulator (CFTR) gene variants on the composition of faecal microbiota, in patients affected by Cystic Fibrosis (CF). CFTR mutations (F508del is the most common) lead to a decreased secretion of chloride/water, and to mucus sticky secretions, in pancreas, respiratory and gastrointestinal tracts. Intestinal manifestations are underestimated in CF, leading to ileum meconium at birth, or small bowel bacterial overgrowth in adult age. Thirty-six CF patients, fasting and under no-antibiotic treatment, were CFTR genotyped on both alleles. Faecal samples were subjected to molecular microbial profiling through Temporal Temperature Gradient Electrophoresis and species-specific PCR. Ecological parameters and multivariate algorithms were employed to find out if CFTR variants could be related to the microbiota structure. Patients were classified by two different criteria: 1) presence/absence of F508del mutation; 2) disease severity in heterozygous and homozygous F508del patients. We found that homozygous-F508del and severe CF patients exhibited an enhanced dysbiotic faecal microbiota composition, even within the CF cohort itself, with higher biodiversity and evenness. We also found, by species-specific PCR, that potentially harmful species (*Escherichia coli* and *Eubacterium bifforme*) were abundant in homozygous-F508del and severe CF patients, while beneficial species (*Faecalibacterium prausnitzii*, *Bifidobacterium* spp., and *Eubacterium limosum*) were reduced. This is the first report that establishes a link among CFTR variants and shifts in faecal microbiota, opening the way to studies that perceive CF as a 'systemic disease', linking the lung and the gut in a joined axis.

The origin of the major cystic fibrosis mutation (delta F508) in European populations.

PMID: 7920636 | Indexed in PubMed

delta F508 is the most frequent cystic fibrosis (CF) mutation and accounts for approximately 70% of CF chromosomes worldwide. Three highly polymorphic microsatellite markers have been used to study the origin and evolution of delta F508 chromosomes in Europe. Haplotype data demonstrate that delta F508 occurred more than 52,000 years ago, in a population genetically distinct from any present European group, and spread throughout Europe in chronologically distinct expansions, which are responsible for the different frequencies of delta F508 in Europe.

An abnormal distribution of delta F508 genotypes in cystic fibrosis patient registries.

PMID: 9599649 | Indexed in PubMed

Delta F508 mutation of the CFTR gene is the most frequent deleterious allele involved in cystic fibrosis (CF). We have studied the distribution of the three genotypes, delta F508/delta F508, delta F508/x, x/x, in the American, Canadian and French data registries concerning CF; "x" represents the non-delta F508 mutations. In the three registries the observed distribution of the three genotypes differs from the expected one, calculated according to the Hardy and Weinberg equilibrium. Three factors could explain this discrepancy: Wahlund's effect, misinterpretation of the molecular diagnosis, or an ascertainment bias in relation with the severity of the disease. This last factor is the most likely.

Frequency of Genotype With Δ F508 Mutation in CFTR Gene Among Iranian Cystic Fibrosis Patients With Pancreatic Insufficiency.

PMID: 26635942 | Indexed in PubMed

Cystic fibrosis (CF) is the most prevalent lethal autosomal recessive disease with a broad spectrum of phenotypes. Mutation of Δ F508 in the CFTR gene is the most important and lethal mutation in CF, which contains 70% of all predisposing mutations for CF worldwide. Determining frequency of genotypes with Δ F508 mutation in CFTR gene, and evaluation of correlation between genotype and phenotype of Iranian patients with CF. Thirty six patients were included in this cross sectional study. Δ F508 mutations in both alleles of the CFTR gene were checked. Among 36 pediatric patients, Δ F508 mutation was detected in 9 (25%) patients; 2 patients were heterozygous, and 7 patients homozygous for this mutation. From overall 72 tracked alleles, 11 (15.2%) were found to have Δ F508 mutations. Differences in prevalence of dyspnea and bronchiectasis were significant in homozygote group, compared with non-mutated group for Δ F508. It seems that more Δ F508 mutated alleles lead to more severe symptoms of CF.

Dysfunction of CFTR bearing the delta F508 mutation.

PMID: 7511616 | Indexed in PubMed

The cystic fibrosis transmembrane conductance regulator (CFTR) is mutated in patients with cystic fibrosis (CF). The most common CF-associated mutation is deletion of phenylalanine at residue 508, CFTR delta F508. When expressed in heterologous cells, CFTR bearing the delta F508 mutation fails to progress through the normal biosynthetic pathway and fails to traffic to the plasma membrane. As a result, CFTR delta F508 is mislocalized and is not present in the apical membrane of primary cultures of airway epithelia. Consequently, the apical membrane of CF airway epithelia is Cl⁻-impermeable, a defect that probably contributes to the pathogenesis of the disease.

Functional activation of the cystic fibrosis trafficking mutant delta F508-CFTR by overexpression.

PMID: 7733303 | Indexed in PubMed

The most common mutation in the gene associated with cystic fibrosis (CF) causes deletion of

phenylalanine at residue 508 (delta F508) of the gene product called CFTR. This mutation results in the synthesis of a variant CFTR protein that is defective in its ability to traffic to the plasma membrane. Because earlier studies showed delta F508-CFTR retains significant phosphorylation-regulated chloride (Cl⁻) channel activity, processes capable of restoring the mislocalized delta F508-CFTR to the correct cellular destination may have therapeutic benefit. Here we report one such process that involves overexpression of the mutant protein and appears to result in the escape of a small amount of delta F508-CFTR to the plasma membrane. In recombinant cells where expression of delta F508-CFTR is controlled by the metallothionein promoter, this effect can be brought about by treatment with sodium butyrate. Although cAMP-activated Cl⁻ channel activity could also be detected in immortalized human airway epithelial cells homozygous for the delta F508 mutation at the single cell level, treatment with butyrate did not generate a measurable cAMP-stimulated Cl⁻ current in polarized monolayers of primary CF airway epithelia. However, the observation that overexpression can effect the presence of recombinant delta F508-CFTR at the plasma membrane suggests that perhaps other butyrate-like compounds that are more potent and more specific for the promoter of the CF gene may be efficacious in alleviating the Cl⁻ channel defect associated with CF.

Identification of revertants for the cystic fibrosis delta F508 mutation using STE6-CFTR chimeras in yeast.

PMID: 7682896 | Indexed in PubMed

Mutations in the gene encoding the cystic fibrosis transmembrane conductance regulator (CFTR) cause cystic fibrosis; the most common mutation is deletion of phenylalanine at position 508 (delta F508). We constructed STE6-CFTR chimeras with portions of the first nucleotide-binding domain (NBD1) of the yeast STE6 a-factor transporter replaced by portions of CFTR NBD1. The chimeras were functional in yeast, but mating efficiency decreased when delta F508 was introduced into NBD1. We isolated two delta F508 revertant mutations (R553M and R553Q) that restored mating; both were located within the CFTR NBD1 sequence. Introduction of these revertant mutations into human CFTR partially corrected the processing and Cl⁻ channel gating defects caused by the delta F508 mutation. These results suggest that the NBD1s of CFTR and STE6 share a similar structure and function and that, in CFTR, the regions containing F508 and R553 interact. They also indicate that the abnormal conformation produced by delta F508 can be partially corrected by additional alterations in the protein.